Quiz 9

Friday, November 6 — RREF and one-step dynamics (Lecture 22)

Last updated: November 3, 2026.

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

Covers Lecture 22 only: reading RREF (pivot / free variables; unique / none / infinitely many) and one-step linear dynamics $x_{t+1} = Ax_t$ (build A, compute x_1). Same-day determinants and Lab 4 are **not** on this quiz. Matrix products / powers (Lecture 23) and Lab 3 are **not** on this quiz.

Schedule

Core

Fluency

C9.1. The RREF of an augmented matrix is

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 3 \\ 0 & 1 & -1 & 4 \\ 0 & 0 & 0 & 0 \end{array}\right].$$

(a) Which variables are pivots? Which are free?
(b) Write the general solution $x = (x_1, x_2, x_3)$ in terms of a parameter t.

C9.2. Classify each RREF as **unique**, **none**, or **infinitely many** solutions. One short reason each.

(a)
$$\left[\begin{array}{cc|c} 1 & 0 & 2 \\ 0 & 1 & -1 \end{array}\right]$$

(b)
$$\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 0 & 0 & 1 \end{array}\right]$$

(c)
$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 1 \\ 0 & 0 & 1 & 4 \end{array}\right]$$

C9.3. From the RREF